INCH-POUND

MIL-C-24643/11C
14 March 1994
SUPERSEDING
MIL-C-24643/11B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSSSF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

* Qualification required.

Construction (watertight)

First - Copper conductor - AWG size 300 MCM, class G *stranding, uncoated (see note 1)*

* Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Ethylene propylene rubber insulation. Minimum average wall thickness 0.075 inch.

Fourth - Reinforcement at manufacturer's option.

* Fifth - Cross-linked polyolefin jacket bonded to the underlying insulation. Minimum average wall thickness 0.075 inch.

Note: 1. The copper shall be hard drawn, having a minimum tensile strength of 67.200 lb/in^2, and minimum elongation of 0.85 percent in 60 inch lengths before stranding. The resistance specified is based on 102.7 percent of the values required for annealed copper.

TABLE I. Details

Military part no. M24643/11	Type and size	Conductor Size AWG	Overall diameter	
			minimum (inches)	maximum (inches)
-01UN	LSSSF-7	300 MCM (class G)	1.020	1.100

AMSC N/A FSC 6145

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EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	0.0388
* Voltage withstand - volts, root mean square, minimum	
Conductor to ground	2500
Insulation resistance - conductor to conductor in water, megohms/1000 feet, minimum	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
* Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - at minus 30 ±2°C, two turns around mandrel with the diameter eight times that of specimen	No damage
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
* Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	125
Heat distortion - percent of unaged, maximum	30

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EXAMINATION AND TESTS (continued): Requirements

Group D:

*	Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
* Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH:

Minimum 1045, maximum 1155 feet

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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